

IPv6 Subnetting Scenario (AI Generated)

Network Prefix: 2001:db8:1000::/48

Subnet 1 (Office LAN) Prefix: 2001:db8:1000:1::/64 Example Host:
2001:db8:1000:1::10

Subnet 2 (Server LAN) Prefix: 2001:db8:1000:2::/64 Example Host:
2001:db8:1000:2::20

Subnet 3 (Guest Wi-Fi) Prefix: 2001:db8:1000:3::/64 Example Host:
2001:db8:1000:3::30

Manual IPv6 Configuration Steps (Windows)

1. Open Control Panel.
2. Go to Network and Sharing Center.
3. Click Change adapter settings.
4. Right-click Ethernet (or Wi-Fi) and select Properties.
5. Open Internet Protocol Version 6 (TCP/IPv6).
6. Select "Use the following IPv6 address".
7. Enter: IPv6 Address : 2001:db8:1000:1::10 Prefix Length: 64 Default Gateway: (leave blank for local testing)
8. Click OK to save.

9. Verify using: ipconfig

Result: The system is configured with the IPv6 address

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Ethernet adapter Ethernet 3:
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Connection-specific DNS Suffix . :
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IPv6 Address. . . . . : 2001:db8:1000:1::10
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IPv6 Address. . . . . : 2409:40c1:403d:d10b:ad5e:ac12:839d:f899
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Temporary IPv6 Address. . . . . : 2409:40c1:403d:d10b:8153:8f9d:aa33:788a
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